

Dylan Pierre

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PROFESSIONAL SUMMARY

Computer Science graduate with experience across backend development, systems programming, machine learning, and software security. Seeking a Software Engineering role focused on building software that is reliable, scalable, performant, and secure.

TECHNICAL PROJECTS

Checkpoint | Go, PostgreSQL, Redis, Nginx, Docker, Git/GitHub | [\[GitHub\]](#)

- Built a concurrent authentication and authorization backend in **Go**, utilizing **PostgreSQL** for persistence, **Redis** for rate limiting and token state, and **Nginx** for redundancy for load balancing horizontally scaled containerized instances with **Docker**.
- Load tested the service under concurrent traffic, achieving **12,100+ requests/second** throughput with a **100% success rate** and **23.3ms p99 tail latency** across **121,000+ requests**.
- Hardened the service against credential theft, token forgery, brute-force attacks, and account enumeration using OWASP-tuned **Argon2id** password hashing, Ed25519-signed **PASETO** tokens, and **Redis**-backed rate limiting.
- Published a standalone **verification library** with integration documentation, enabling third-party services to authenticate against the service without depending on the underlying token protocol.

Backbone | Python, PyTorch, Machine Learning, Computer Vision, CT | [\[Link\]](#)

- Designed and implemented a machine learning & computer vision pipeline for cervical spine fracture detection in CT scan imaging, utilizing **PyTorch/Ultralytics** for YOLOv8 object detection, **pydicom** and **OpenCV** for DICOM preprocessing, and the RSNA 2022 CT dataset for training data.
- Processed a balanced training set of **14,434 labeled CT slices** (11,547 train / 2,887 validation) and fine-tuned an **11.1M-parameter YOLOv8s** model capable of screening a full CT study in **under 2 seconds** on GPU.
- Applied Hounsfield Unit windowing (290–1700 HU) to isolate bone structures, balanced the dataset 1:1 to counter severe class imbalance, and split data at the patient/study level to prevent leakage between training and validation sets.
- Collaborated on an **AGILE** team under a fast-paced timeline, running weekly stand-ups and delegating pipeline components to deliver a working system.

C-Like Language Interpreter | C++, Interpreters, CMake | [\[GitHub\]](#)

- Built a full language interpreter in C++, implementing the tokenizer, recursive-descent parser, concrete syntax tree, abstract syntax tree, symbol table, and evaluator from scratch with **zero external dependencies**.
- Correctly interprets programs containing typed variables (**int**, **char**, **bool**), arrays, functions and procedures, **if/else**, **for**, and **while** control flow, arithmetic/relational/logical expressions with proper precedence, and formatted output, supported by **scope-aware redeclaration checks** and **line-numbered error diagnostics**.
- Modeled lexical analysis as an explicit **28-state deterministic finite automaton** and validated the **4,600+ line** implementation against **25 test programs** spanning each component of the interpreter pipeline.

EXPERIENCE

DataAnnotation

May 2024 – Present

Code Quality Analyst

Remote

- Utilized **AI coding tools**, including **Claude Code** and **GitHub Copilot**, to make modifications and extend functionality across existing codebases in **Python**, **JavaScript**, **C++**, and **Java**.
- Reviewed and debugged **LLM-generated code** against technical specifications, evaluating **correctness**, **API usage**, and **edge case handling** to drive model training improvements.

EDUCATION

Sonoma State University

3.5 GPA

Bachelor of Science in Computer Science

2025

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture, Algorithm Analysis, Operating Systems, Database Systems, Computer Networks, Programming Languages, Software Engineering, Machine Learning, Artificial Intelligence

CERTIFICATIONS

CompTIA Security+ (April 2026), **Google Cybersecurity Professional**